

RUSE, Bulgaria

WMO: 155350

Lat: 43.850N

Lon: 25.950E

Elev: 45

StdP: 100.79

Time Zone: 2.00 (EUE)

Period: 94-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.6	-9.0	-13.5	1.2	-11.2	-10.7	1.5	-8.6	17.1	0.9	15.8	2.0	2.5	50	0.494

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.6	35.5	22.4	33.6	21.9	32.0	21.3	23.9	32.3	23.0	31.2	22.2	30.1	3.2	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.2	15.9	28.6	20.2	15.0	27.1	19.5	14.3	26.0	71.7	32.3	68.2	31.1	65.2	30.2	28.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.4	12.2	9.9	DB	-14.5	39.0	3.8	2.2	-17.2	40.6	-19.5	41.9	-21.6	43.2	-24.3	44.8	(4)
(5)			WB	-14.7	26.0	3.8	1.5	-17.4	27.1	-19.7	28.0	-21.8	28.8	-24.6	29.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	13.1	0.0	2.6	7.9	13.5	18.9	22.7	24.8	24.7	19.5	13.3	7.1	1.3		
	DBStd	9.76	5.18	5.64	4.87	3.83	3.55	3.28	2.93	3.04	3.65	4.50	4.54	5.02		
	HDD10.0	1039	312	210	99	14	0	0	0	0	0	20	111	273		
	HDD18.3	2588	569	439	323	151	36	5	1	1	29	166	338	529		
	CDD10.0	2164	1	4	34	118	278	380	459	457	285	122	23	2		
	CDD18.3	669	0	0	1	4	55	134	202	199	64	9	1	0		
	CDH23.3	6532	0	0	4	51	497	1288	2089	2018	514	71	1	0		
	CDH26.7	2763	0	0	0	12	143	521	974	938	162	14	0	0		
(14) Wind	WSAvg	3.4	3.3	3.8	4.2	4.0	3.6	3.4	3.3	3.0	3.2	2.9	3.1	3.3		
(15) Precipitation	PrecAvg	614	43	33	43	46	67	66	61	46	61	52	46	50		
	PrecMax	994	114	104	105	121	174	168	201	134	198	124	115	165		
	PrecMin	334	3	1	2	3	16	12	3	3	1	0	1	2		
	PrecStd	155	25	21	24	24	41	35	50	38	51	33	28	37		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.4	18.6	23.1	28.2	32.7	36.0	38.5	38.2	33.4	28.6	21.6	16.5		
		MCWB	9.5	11.5	13.2	17.8	20.6	22.9	22.8	22.7	20.5	19.0	14.9	11.6		
	2%	DB	11.6	15.4	20.7	24.3	30.1	33.5	35.8	35.4	30.5	25.2	19.0	12.6		
		MCWB	8.0	10.0	12.2	15.6	19.6	22.3	22.2	22.6	19.7	17.8	14.0	9.4		
	5%	DB	9.0	12.8	18.2	22.1	28.1	31.6	33.6	33.4	28.3	22.3	15.9	9.6		
		MCWB	6.6	8.7	11.4	14.4	18.8	21.7	22.0	22.0	19.1	16.3	12.5	7.4		
	10%	DB	6.6	10.0	15.8	20.2	26.2	29.8	31.7	31.6	26.3	20.0	13.6	7.5		
		MCWB	4.8	7.0	10.0	13.2	18.0	20.8	21.3	21.2	18.3	15.0	11.0	6.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.0	12.0	14.6	18.6	22.5	24.2	24.7	25.3	22.1	20.0	15.6	11.7		
		MCDB	13.4	17.1	20.7	26.8	29.7	32.5	34.4	33.9	29.1	27.1	20.6	15.8		
	2%	WB	8.3	10.3	12.7	16.4	20.7	23.2	23.6	24.1	20.7	18.2	14.2	9.6		
		MCDB	11.0	14.9	19.4	23.1	28.1	31.7	32.6	32.0	28.7	24.1	18.2	12.3		
	5%	WB	6.6	8.8	11.6	15.1	19.5	22.2	22.8	22.9	19.7	16.8	12.8	7.7		
		MCDB	8.8	12.4	17.7	20.8	26.3	30.3	31.2	30.8	26.8	21.5	15.7	9.3		
	10%	WB	4.9	7.1	10.3	14.0	18.5	21.2	22.1	21.9	18.7	15.5	11.2	6.1		
		MCDB	6.5	9.9	15.5	19.0	24.9	28.6	29.9	29.9	24.9	19.5	13.3	7.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.4	7.4	9.3	10.2	11.2	11.4	11.6	11.6	10.4	9.2	6.9	5.5		
		MCDBR	9.0	12.1	13.2	13.5	14.3	14.1	14.7	14.4	13.8	12.8	11.0	9.2		
	5% WB	MCWBR	6.2	7.6	7.3	6.8	6.6	5.8	5.5	5.4	5.8	6.8	6.8	6.4		
		MCWBR	8.5	11.4	12.5	11.9	12.8	13.2	12.9	12.7	12.3	11.9	10.1	8.7		
(40) Clear-Sky Solar Irradiance	taub		0.351	0.384	0.421	0.461	0.435	0.444	0.453	0.467	0.433	0.406	0.377	0.355		
	taud		2.263	2.189	2.139	2.051	2.201	2.211	2.192	2.151	2.211	2.261	2.292	2.277		
	Ebn at Noon		748	786	806	804	840	832	818	791	786	753	711	702		
	Edh at Noon		88	113	135	159	141	140	141	143	124	104	86	80		
(44) All-Sky Solar Radiation	RadAvg		1.46	2.25	3.41	4.71	5.85	6.38	6.48	5.80	4.22	2.73	1.56	1.20		
	RadStd		0.22	0.31	0.33	0.50	0.46	0.46	0.44	0.33	0.33	0.30	0.21	0.17		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (52 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		+0.45	N/A	N/A	N/A	N/A	N/A	-119	+63	N/A

Nomenclature: See separate page